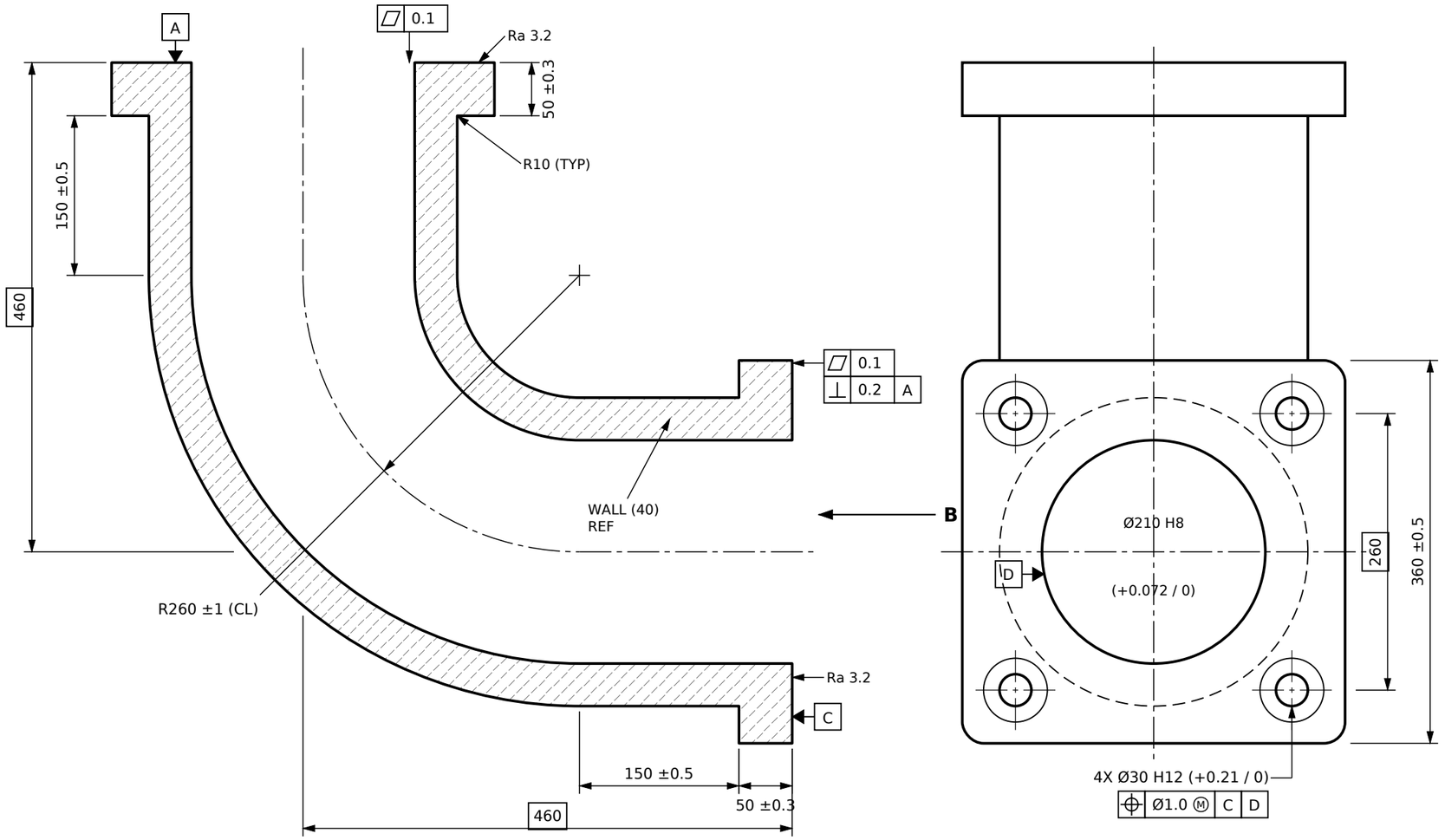
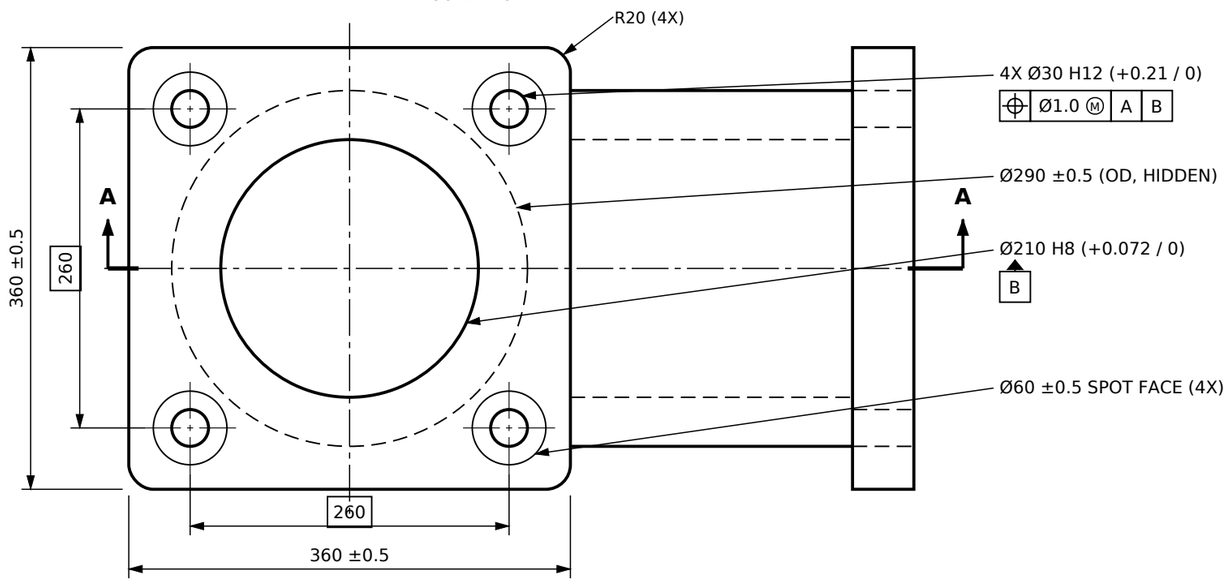


SECTION A-A

SCALE 1:5

VIEW B — FROM FLANGE 2 FACE

SCALE 1:5



VIEW C — FROM FLANGE 1 FACE

SCALE 1:5

TOLERANCE & GD&T SUMMARY

DATUM A	Flange 1 mating face
DATUM B	Ø210 bore axis, flange 1
DATUM C	Flange 2 mating face
DATUM D	Ø210 bore axis, flange 2
Flange faces (A, C)	Flatness 0.1
Flange 2 face (C)	Perpendicularity 0.2 to A
Bores Ø210 (B, D)	H8 (+0.072 / 0)
4X Ø30 holes per flange	H12 (+0.21 / 0)
Position, flange 1	Ø1.0 (M) | A | B
Position, flange 2	Ø1.0 (M) | C | D
Ø60 spot face	±0.5
Flange thickness 50	±0.3
Flange width 360	±0.5
Straight length 150	±0.5
Bend radius R260	±1.0 (centreline)
Pipe OD Ø290	±0.5
Wall thickness 40	Reference only
Boxed dims (260, 460)	Basic dimensions
Surface finish	Ra 3.2 flange faces
	Ra 12.5 elsewhere
General tolerance	ISO 2768-mK

NOTES

1. All dimensions in mm.
2. Boxed dimensions are BASIC.
3. Unspecified tolerances: ISO 2768-mK.
4. Break sharp edges 0.5 max.
5. Corner radii R20, fillets R10 (as modelled).
6. Holes and spot faces are not cut by A-A.
7. Bolt assumed M27; Ø30 = medium clearance.
8. Floating fastener: T = H − F = 30 − 27
   = 3.0 max; Ø1.0 chosen for alignment.
9. Material: AISI 1020 plain carbon steel.

REFERENCE DATA (NOT FOR INSPECTION)

Bolt circle Ø367.7 (REF)
Hole centre to bore axis 183.8 (REF)
Edge distance 50 (REF)
Centreline arc length 408.4 (REF)
Min. wall at 1 MPa ≈ 0.9 mm (design check)

TITLE

90° FLANGED PIPE ELBOW

DWG NO.	SCALE	SIZE	SHEET
MODEL01 DRAFTING	**1:5**	**A3**	**1 OF 1**
MATERIAL	WEIGHT	FINISH	UNITS
AISI 1020	**≈ 247 kg (CALC.)**	**Ra 3.2 / Ra 12.5**	**mm**
DRAWN	DATE	TOLERANCE	REV
A. K. PANDEY	**28-10-2026**	**ISO 2768-mK**	**A**